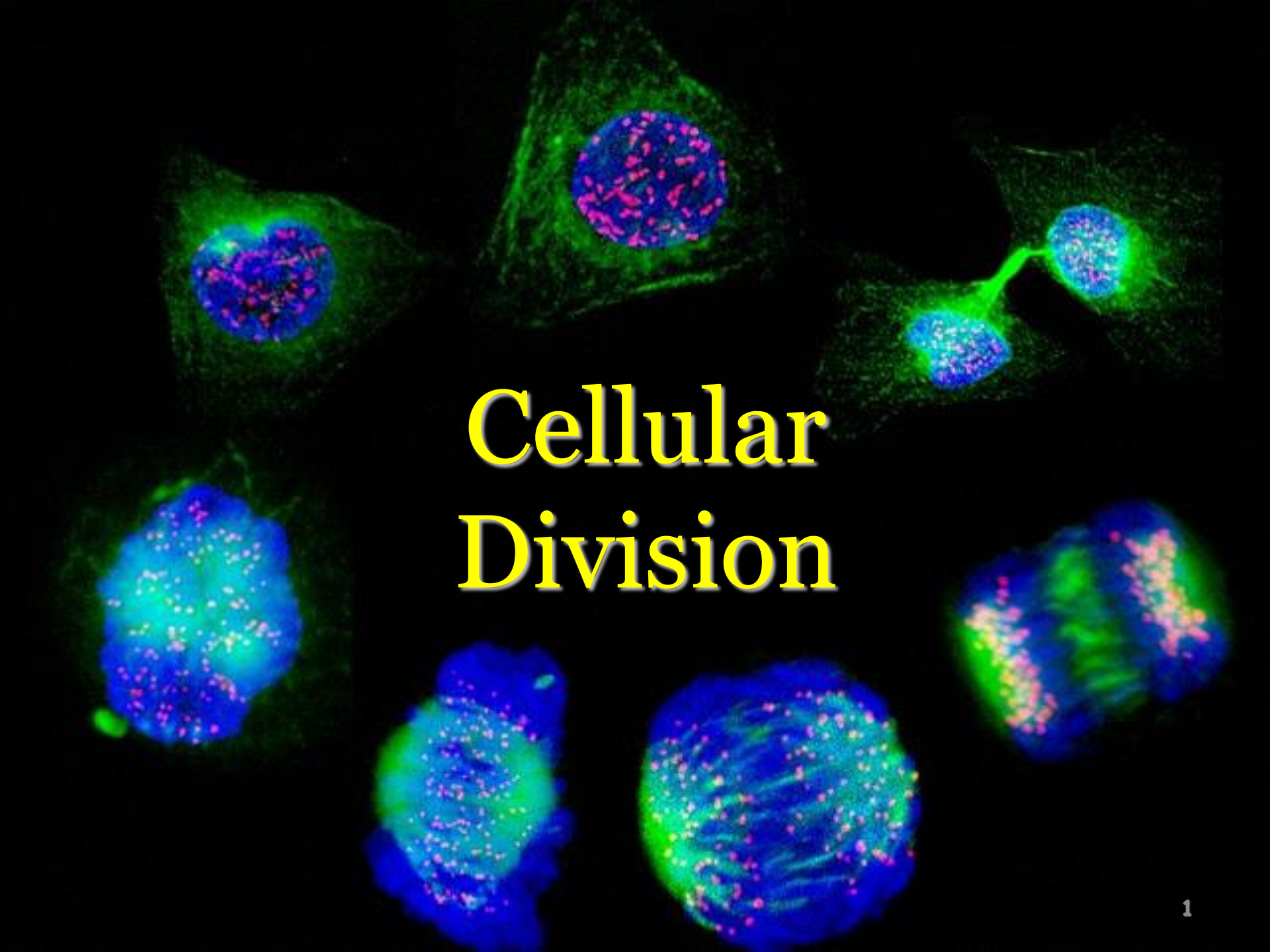


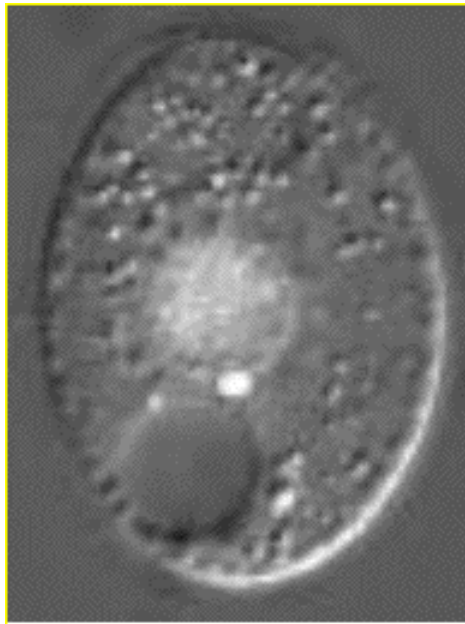
The background of the slide features several fluorescence microscopy images of cells. The cells are stained with a green dye that highlights the cytoskeleton and a blue/purple dye that stains the nuclei. Some cells show distinct chromosomal structures, indicating they are in the process of mitosis or meiosis. One cell in the upper right shows a clear spindle apparatus. Another cell in the lower right shows a more complex, possibly multinucleated or highly condensed chromatin state. The overall image is dark, with the glowing green and blue/purple structures providing the primary visual information.

Cellular Division

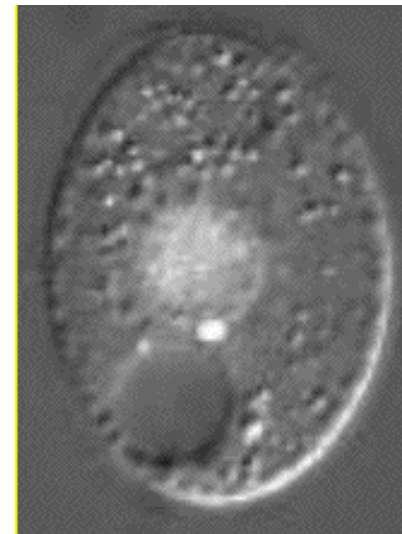
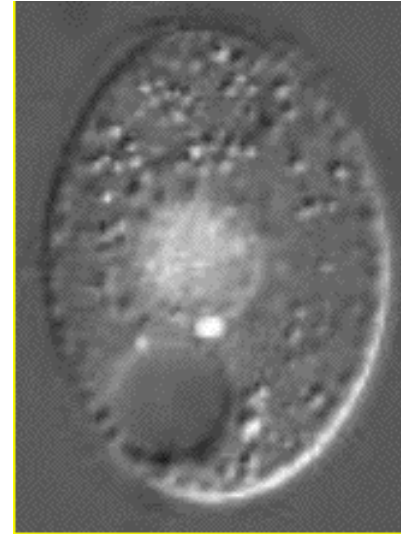
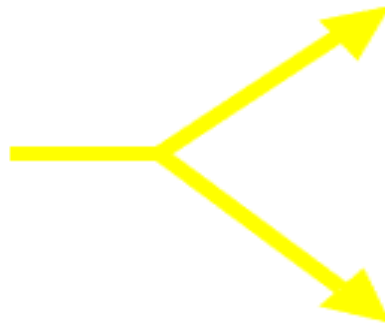
Cell Division

- All cells are derived from **pre-existing** cells
- New cells are produced for **growth** and to replace **damaged or old cells**
- Differs in **prokaryotes** (bacteria) and **eukaryotes** (protists, fungi, plants, & animals)
- The **instructions for making cell parts** are encoded in the **DNA**, so each new cell must get a **complete set of the DNA molecules**

Identical Daughter Cells



Parent Cell

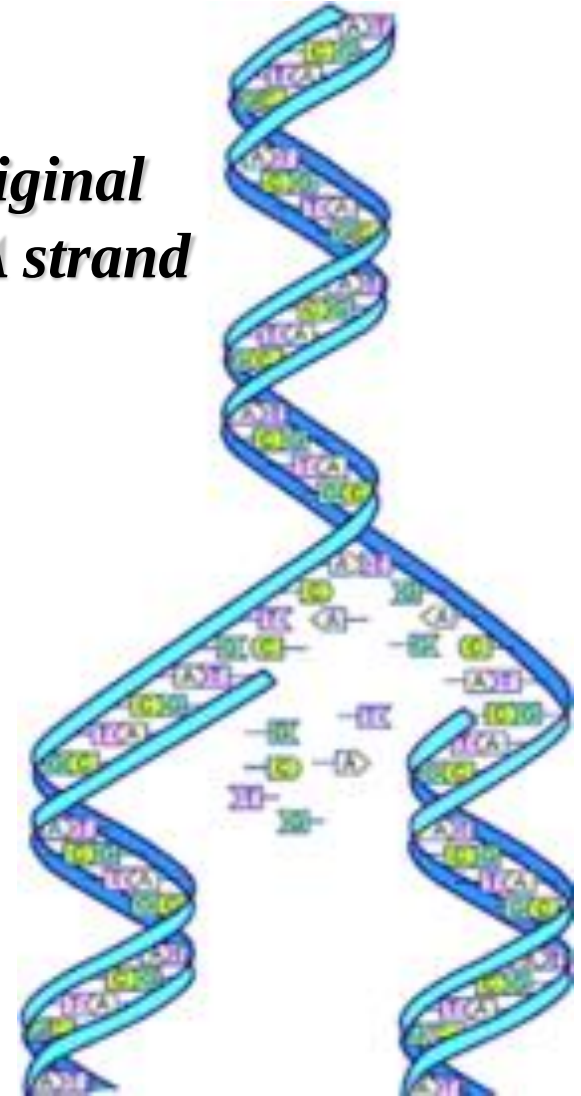


*Two
identical
daughter
cells*

DNA Replication

- ✓ DNA must be copied or **replicated** before cell division
- ✓ Each new cell will then have an **identical copy** of the DNA

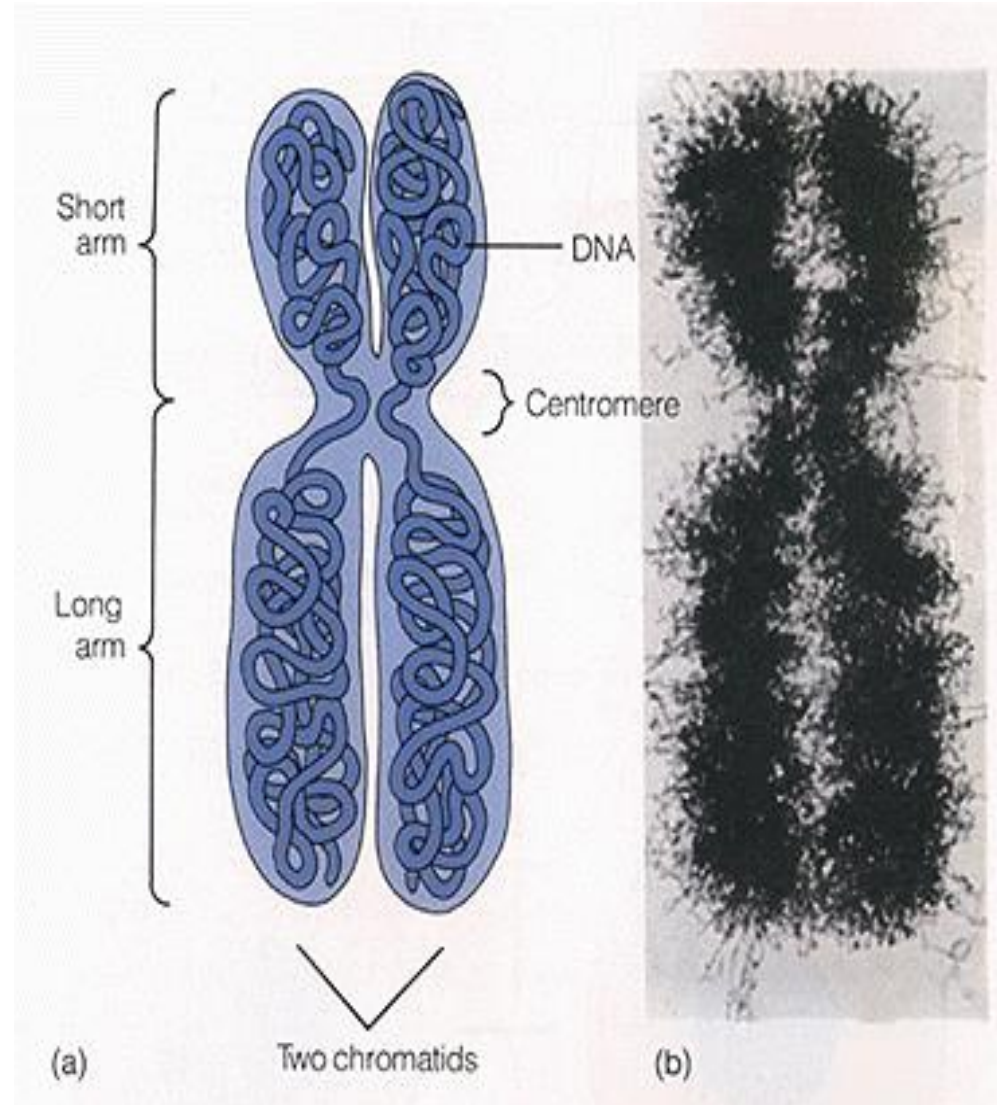
*Original
DNA strand*



*Two new, identical
DNA strands*

Chromosomes in Dividing Cells

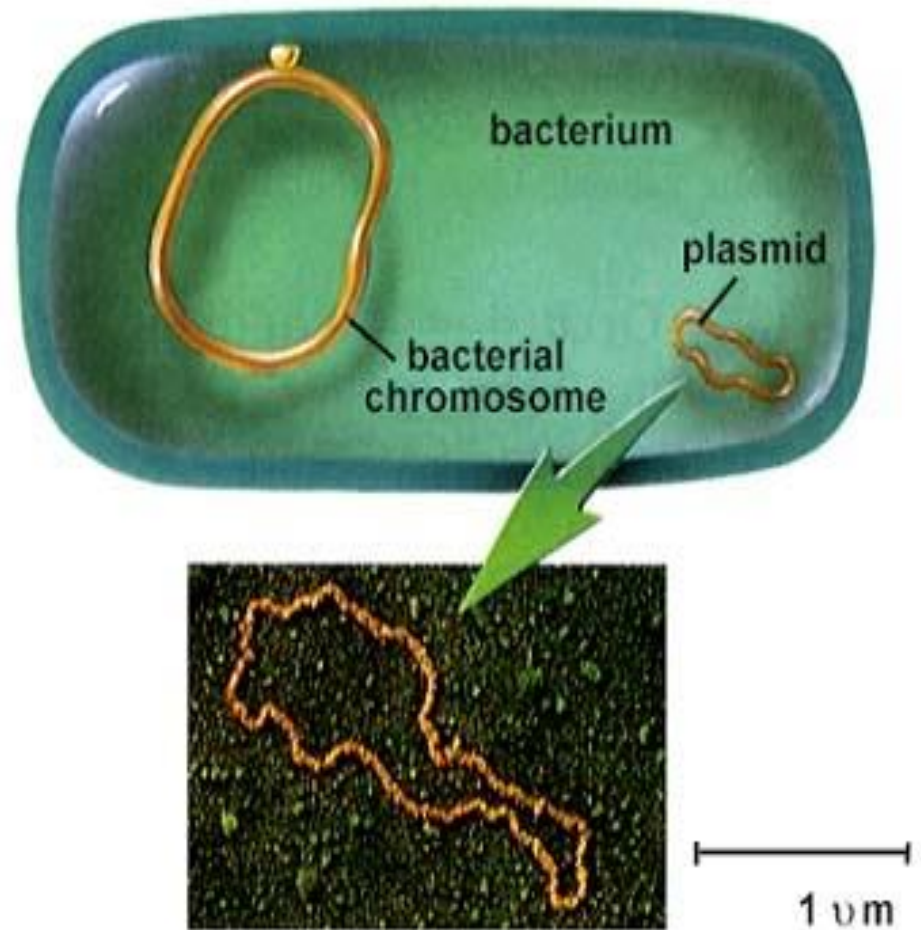
✓ Duplicated chromosomes are called **chromatids** & are held together by the **centromere**



Called Sister Chromatids

Prokaryotic Chromosome

- ✓ The DNA of prokaryotes (bacteria) is **one, circular chromosome** attached to the inside of the cell membrane
- ✓ **Plasmids** are extrachromosomal circular DNA



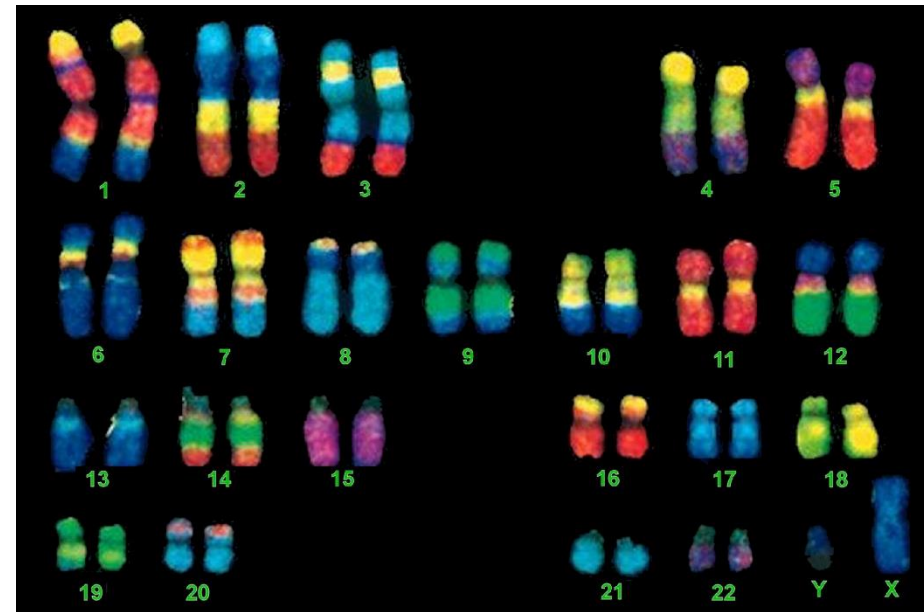
Eukaryotic Chromosomes

- ✓ All **eukaryotic** cells store genetic information in chromosomes
- ✓ Most eukaryotes have between **10 and 50 chromosomes** in their body cells
- ✓ **Human body cells** have **46** chromosomes



Karyotype

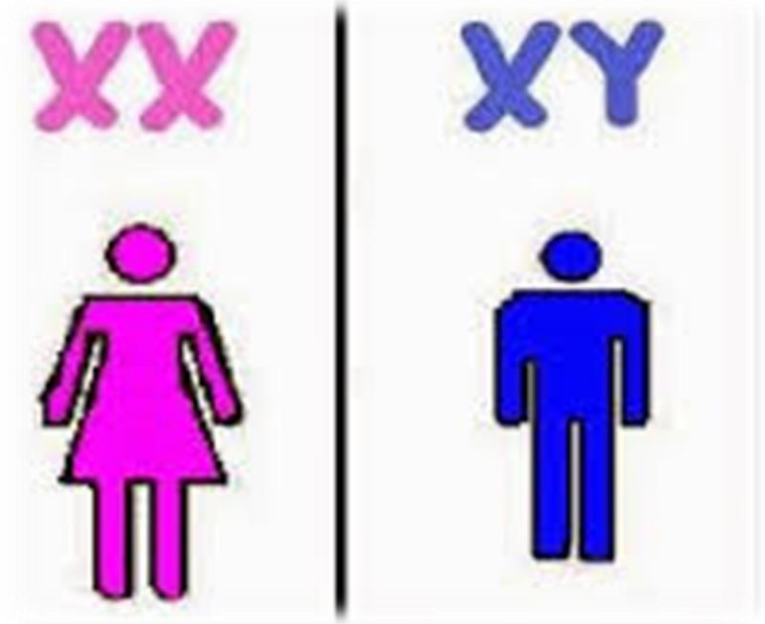
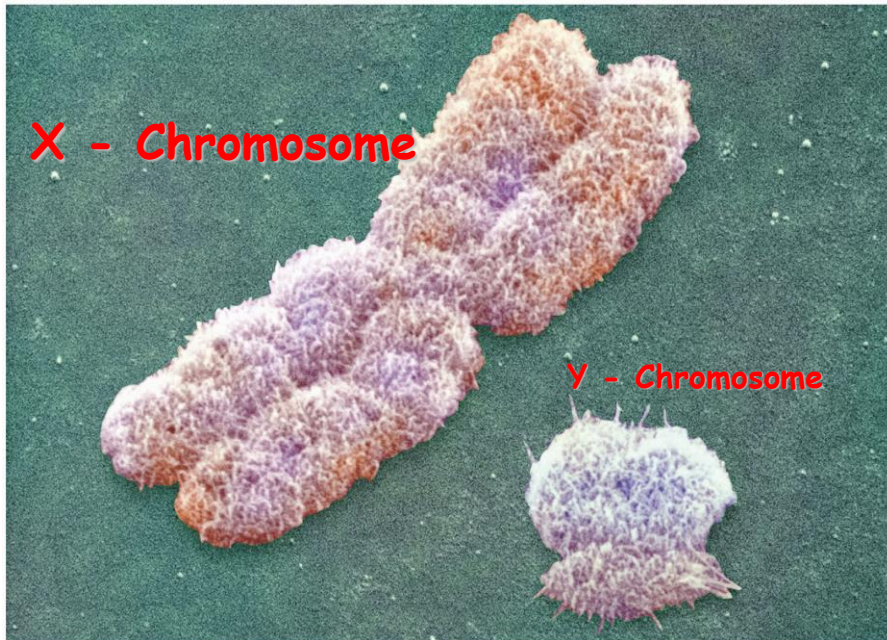
- ✓ A **picture** of the chromosomes from a human cell **arranged in pairs by size**
- ✓ First 22 pairs are called **autosomes**
- ✓ Last two are the **sex chromosomes**
- ✓ **XX** female or **XY** male

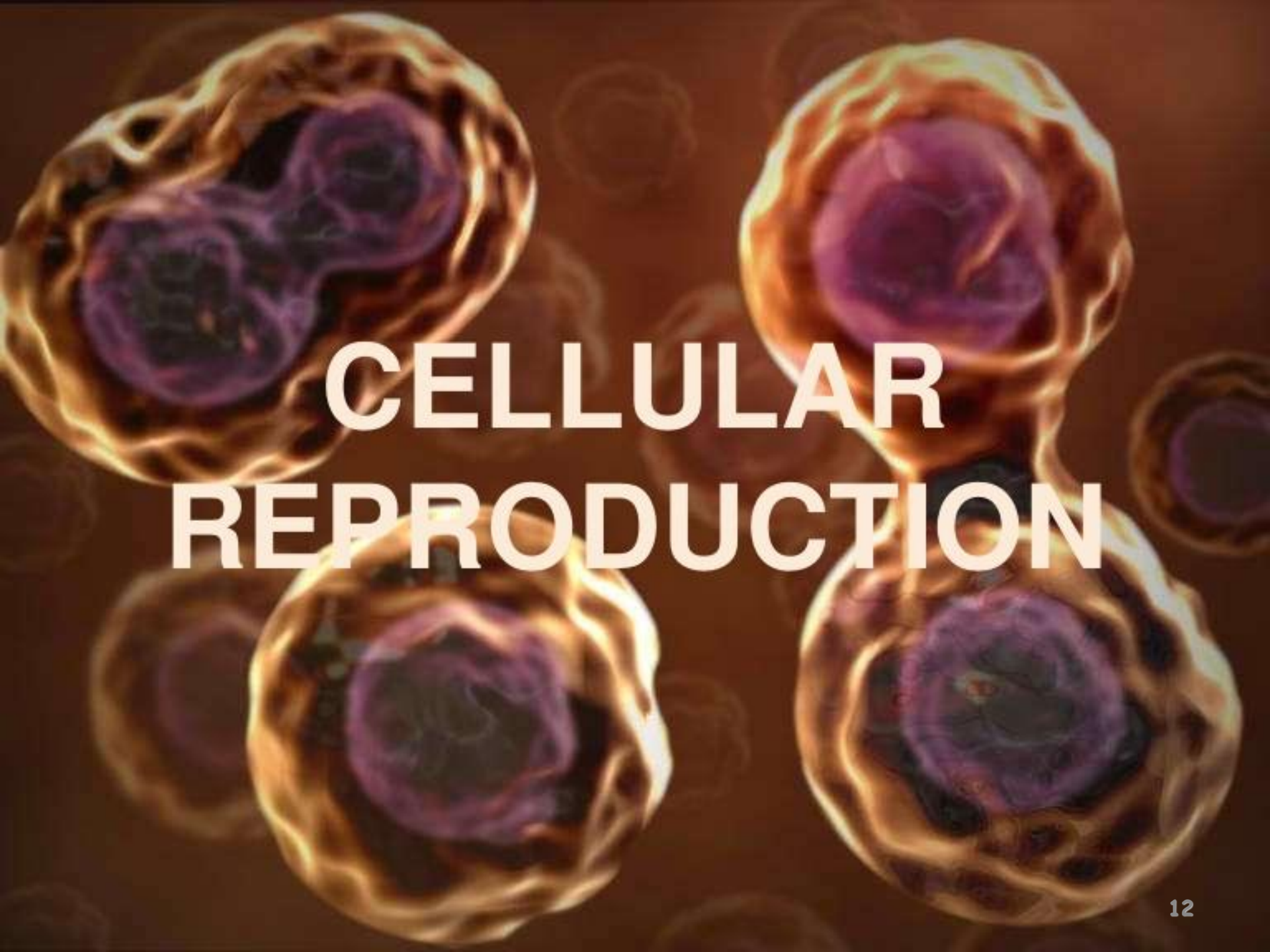




Boy or Girl?

The Y Chromosome Decides



A microscopic image showing several cells with prominent purple nuclei and orange, textured cytoplasm. The cells are arranged in a cluster, with some showing signs of division or budding. The background is a dark, brownish-orange color.

CELLULAR REPRODUCTION

Types of Cell Reproduction

1. Asexual reproduction involves a single cell dividing to make 2 new, identical daughter cells

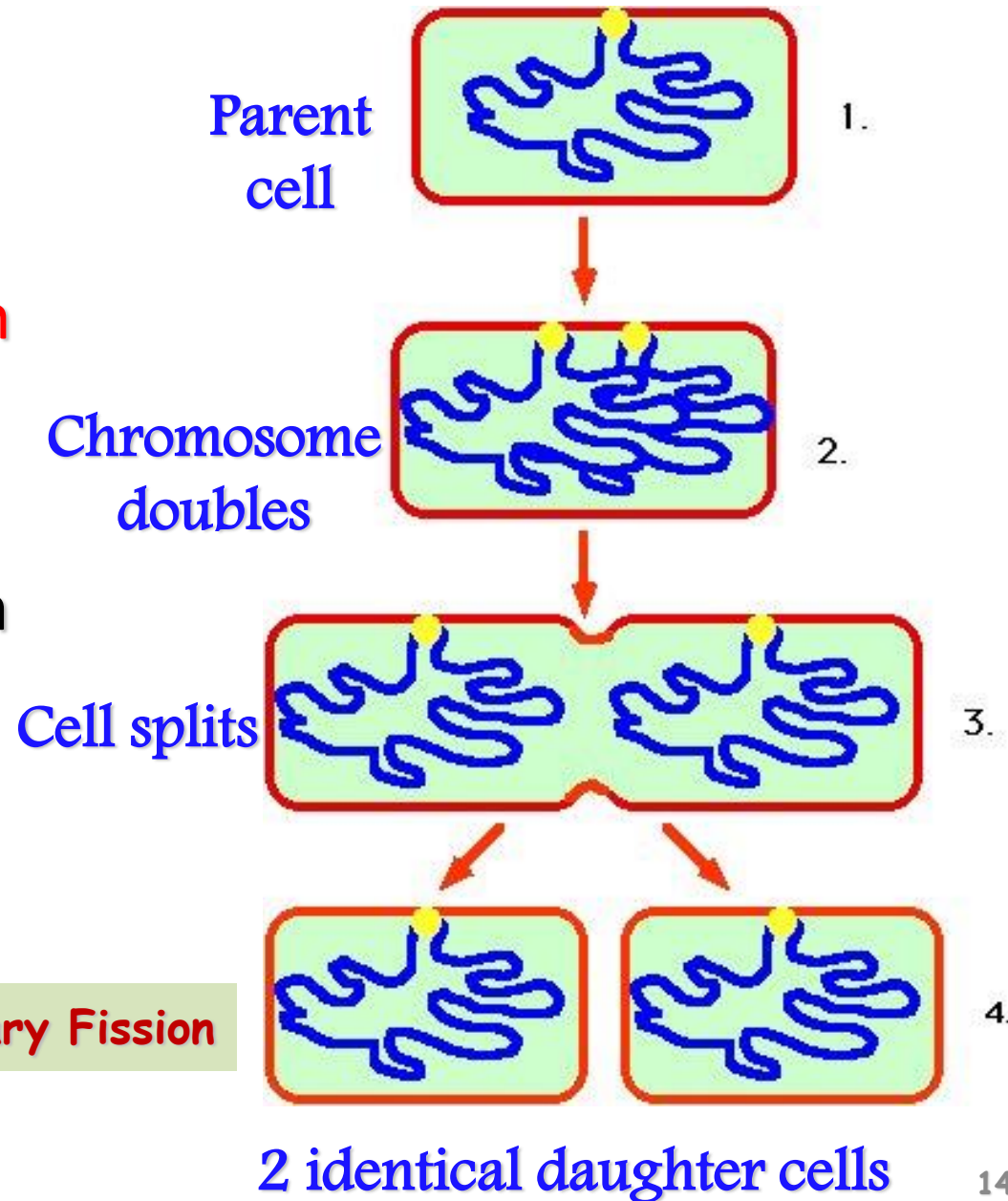
✓ Mitosis & binary fission are examples of asexual reproduction

2. Sexual reproduction involves two cells (egg & sperm) joining to make a new cell (zygote) that is NOT identical to the original cells

✓ Meiosis is an example

Cell Division in Prokaryotes

- ✓ Prokaryotes such as **bacteria** divide into 2 identical cells by the process of **binary fission**
- ✓ **Single chromosome makes a copy of itself**
- ✓ Cell wall forms between the chromosomes dividing the cell



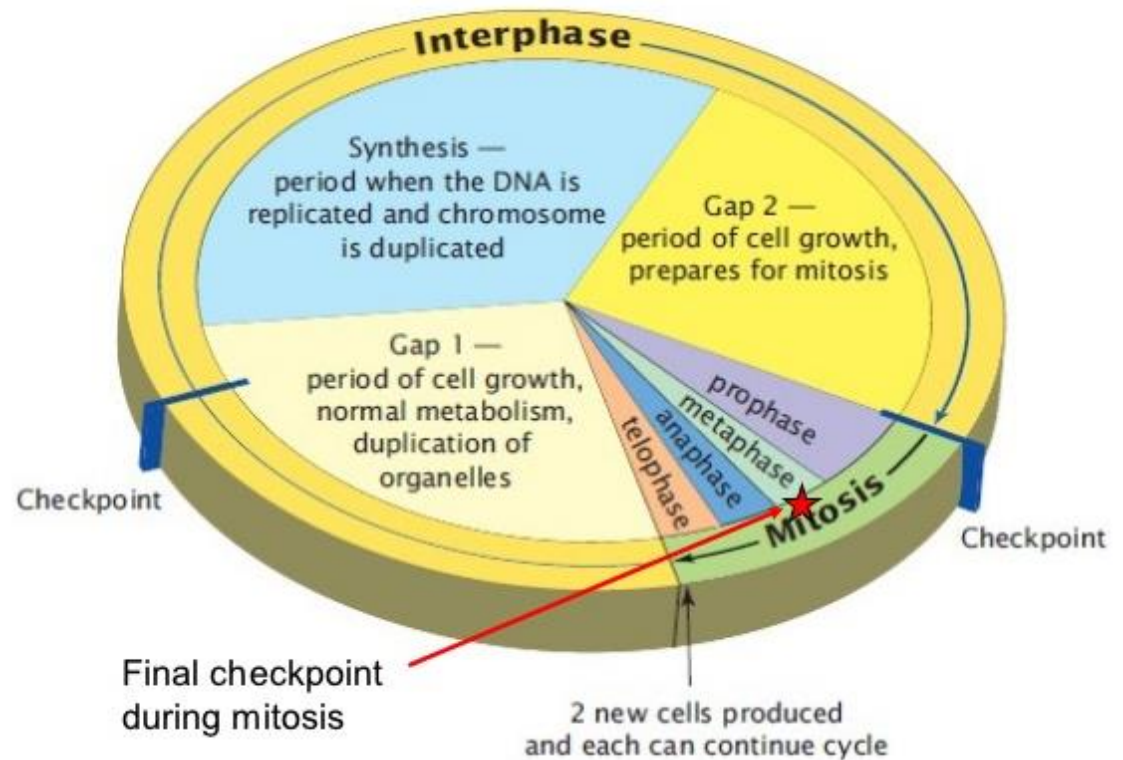
Prokaryotic Cell Undergoing Binary Fission

Five Phases of the Cell Cycle

- ✓ G_1 - primary growth phase
- ✓ S – synthesis; DNA replicated
- ✓ G_2 - secondary growth phase

collectively these 3 stages are called interphase

- ✓ M - mitosis
- ✓ C - cytokinesis



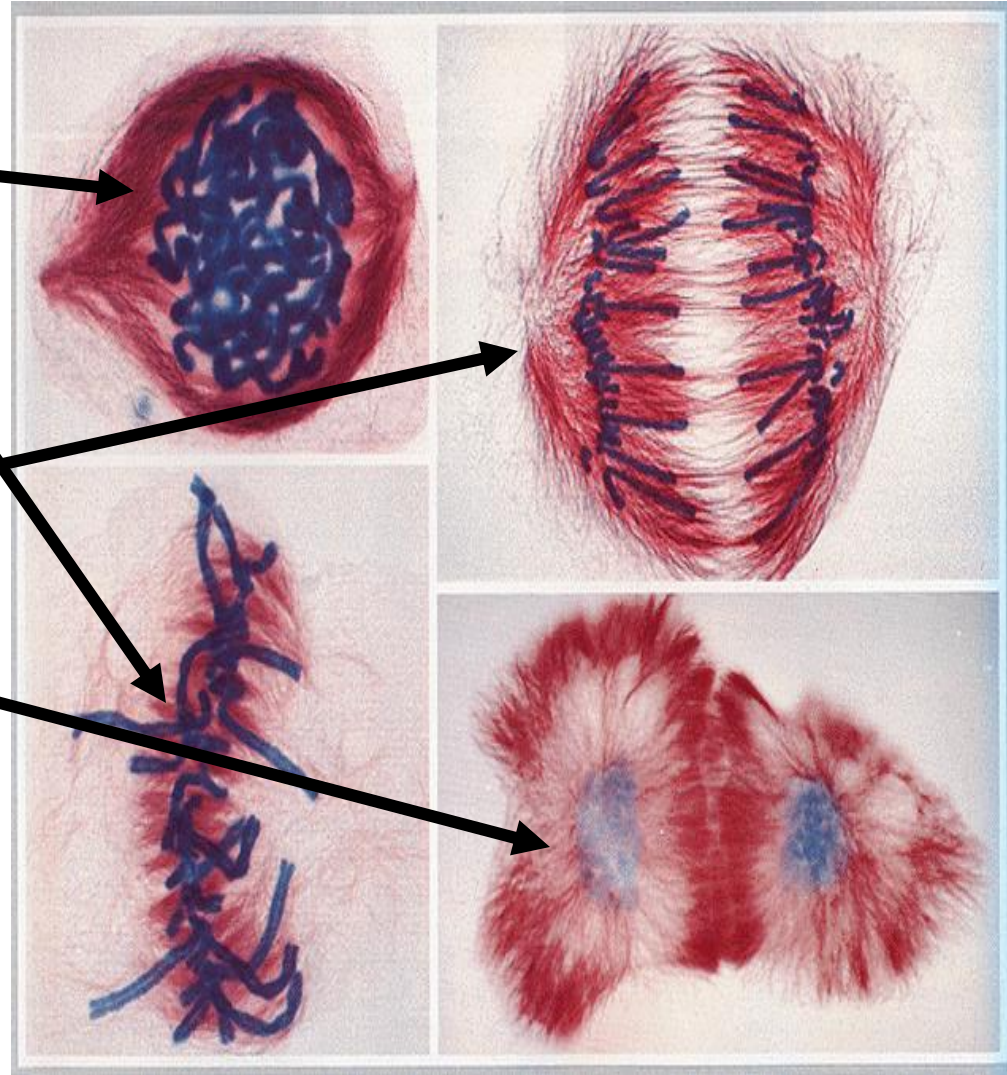
Stages of mitosis

✓ 1. Prophase

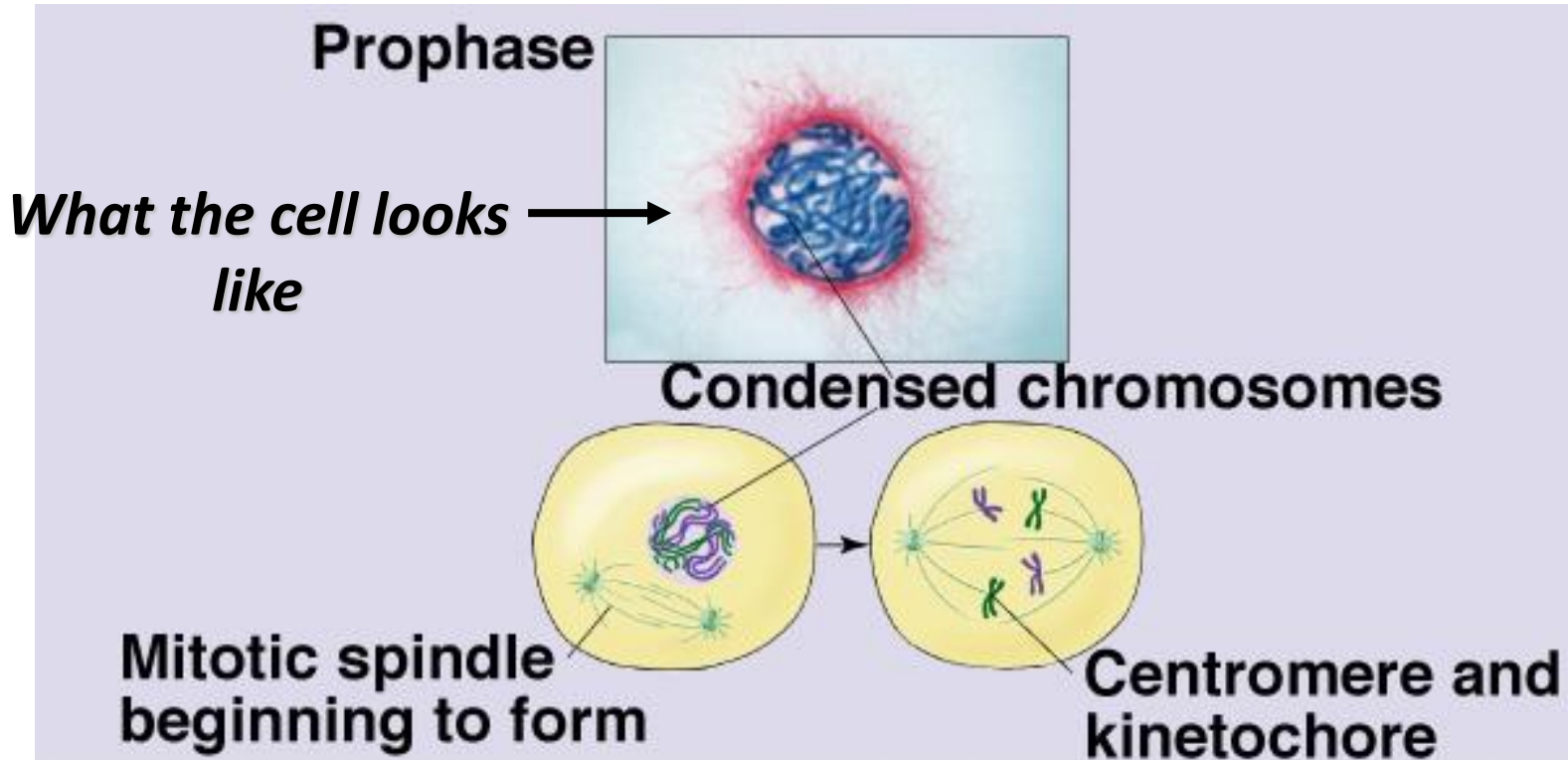
✓ 2. Metaphase

✓ 3. Anaphase

✓ 4. Telophase



1. Prophase

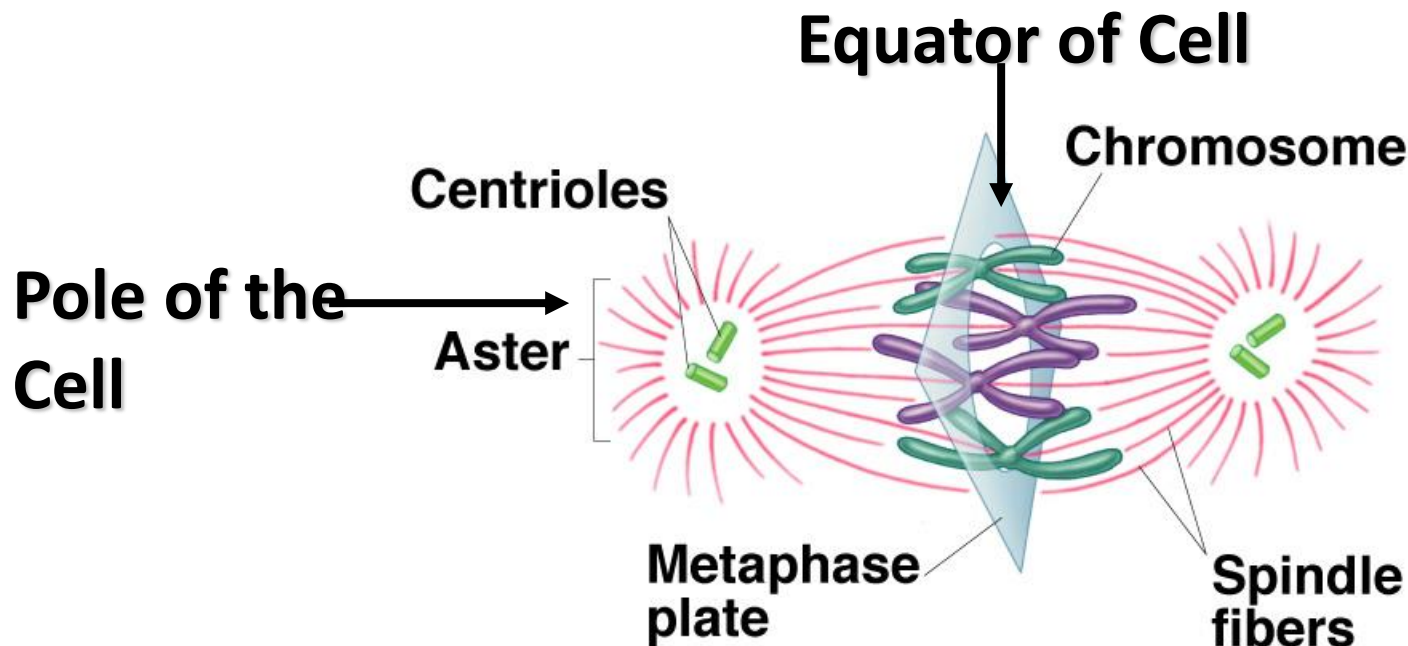


- Nuclear membrane disintegrates, and nucleolus disappears
- Chromosomes condense
- Mitotic spindle begins to form and is complete at the end of prophase
- Kinetochores begin to mature and attach to spindle

What's happening

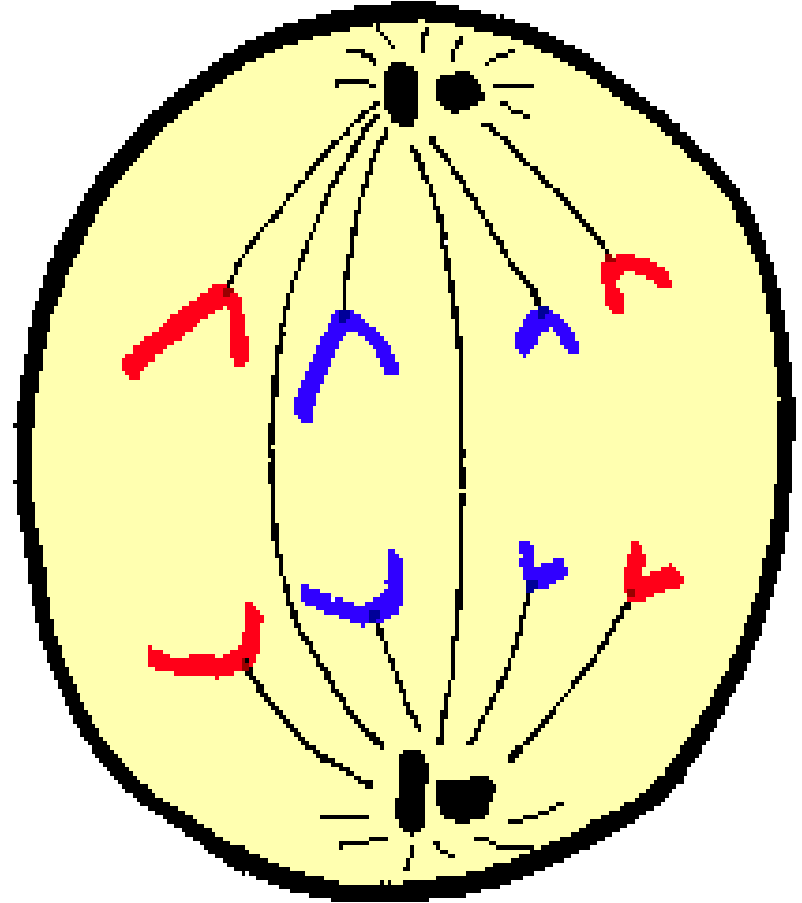
2. Metaphase

- ✓ Chromosomes, attached to the **kinetochore fibers**, move to the center of the cell
- ✓ **Chromosomes are now lined up at the equator**



3. Anaphase

- ✓ Occurs **rapidly**
- ✓ **Sister chromatids** are pulled apart to **opposite poles** of the cell by kinetochore fibers



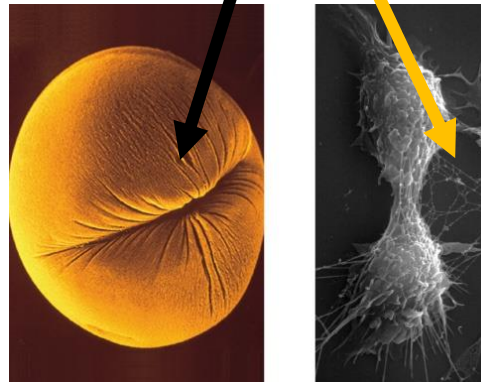
4. Telophase

- ✓ Sister chromatids at **opposite poles**
- ✓ Spindle **disassembles**
- ✓ **Nuclear envelope** forms around each set of sister chromatids
- ✓ **Nucleolus** reappears
- ✓ **CYTOKINESIS** occurs
- ✓ Chromosomes reappear as **chromatin**

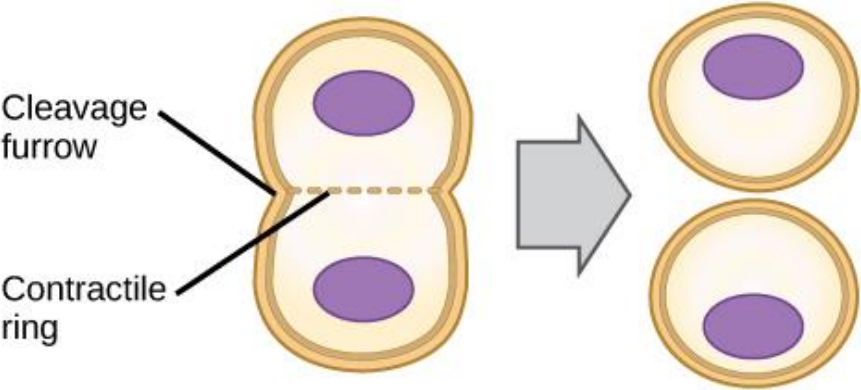
Cytokinesis

- ✓ Means division of the cytoplasm
- ✓ Division of cell into two, identical halves called daughter cells
- ✓ In plant cells, cell plate forms at the equator to divide cell
- ✓ In animal cells, cleavage furrow forms to split cell

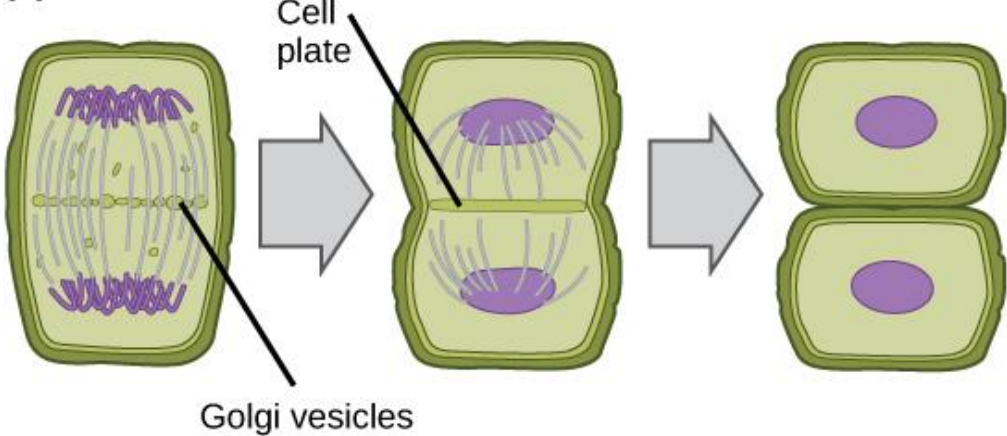
Cleavage furrow in animal cell



(a) Animal cell

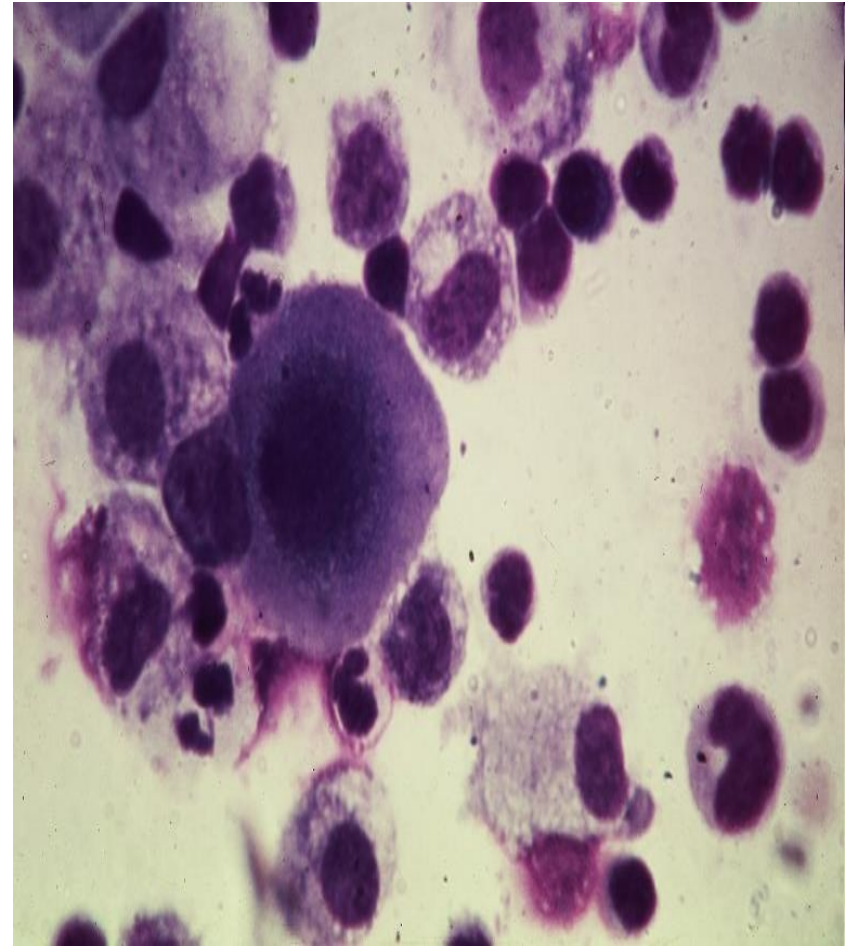


(b) Plant cell



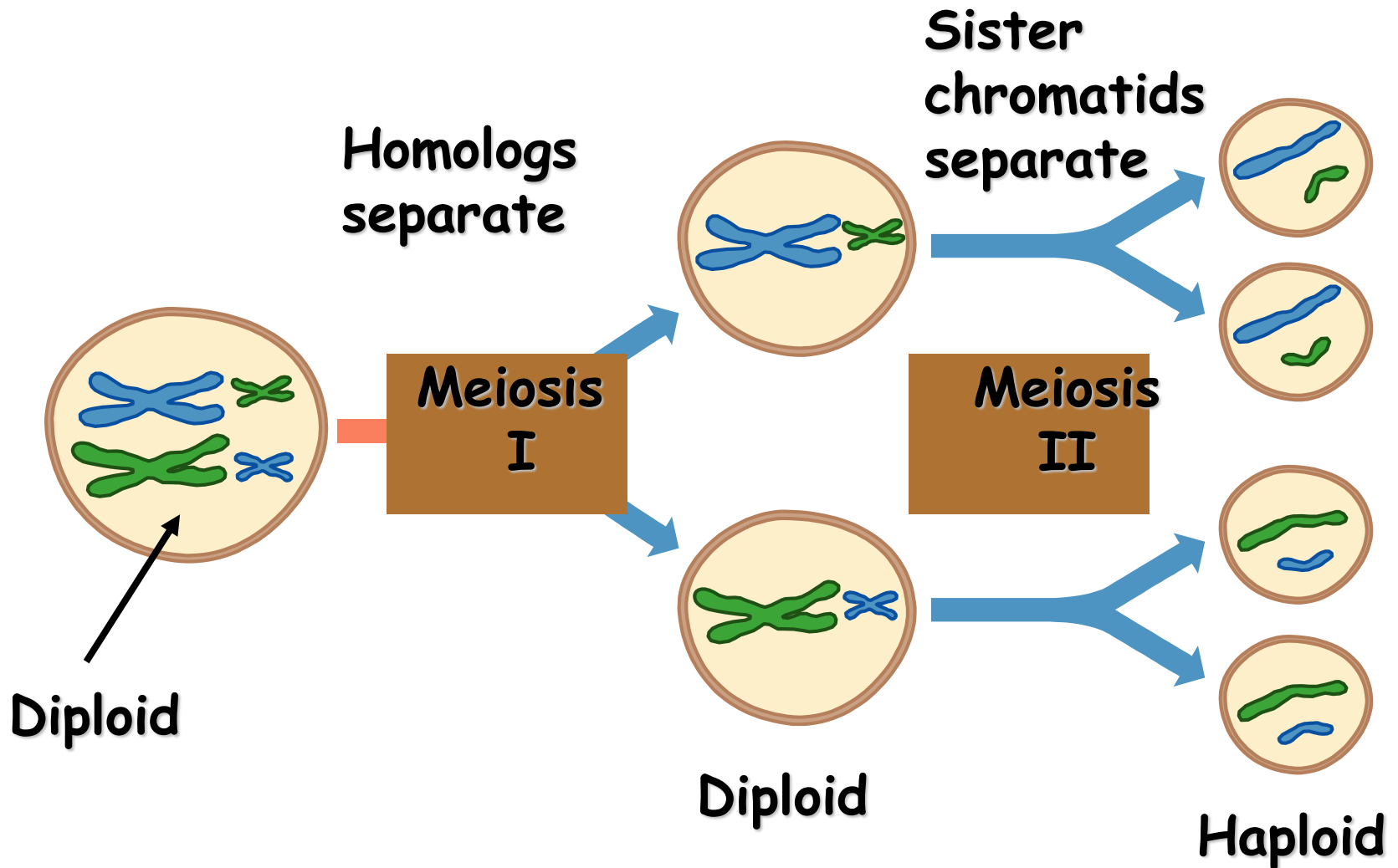
Uncontrolled Mitosis

- ✓ If mitosis is not controlled, unlimited cell division occurs causing cancerous tumors
- ✓ **Oncogenes** are special proteins that increase the chance that a normal cell develops into a tumor cell

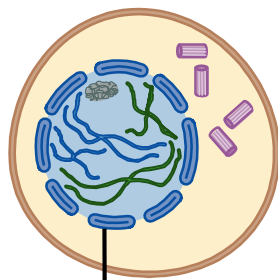


Cancer cells

Meiosis: Two Part Cell Division

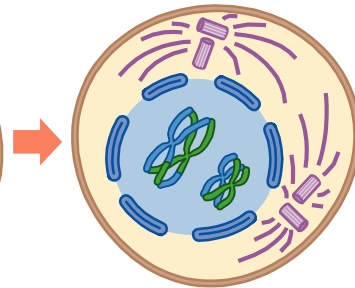


Meiosis I: Reduction Division

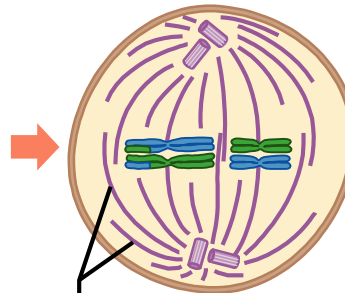


Nucleus

Early
Prophase I
(Chromosome
number doubled)
Crossing over
occurs

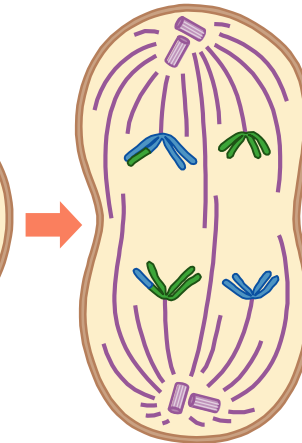


Late
Prophase I
Spindle
forms.
Nuclear
envelope
fragments

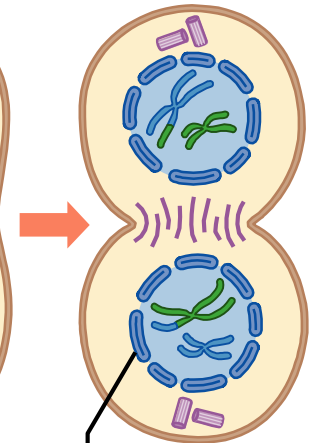


Spindle
fibers

Metaphase
I



Anaphase
I



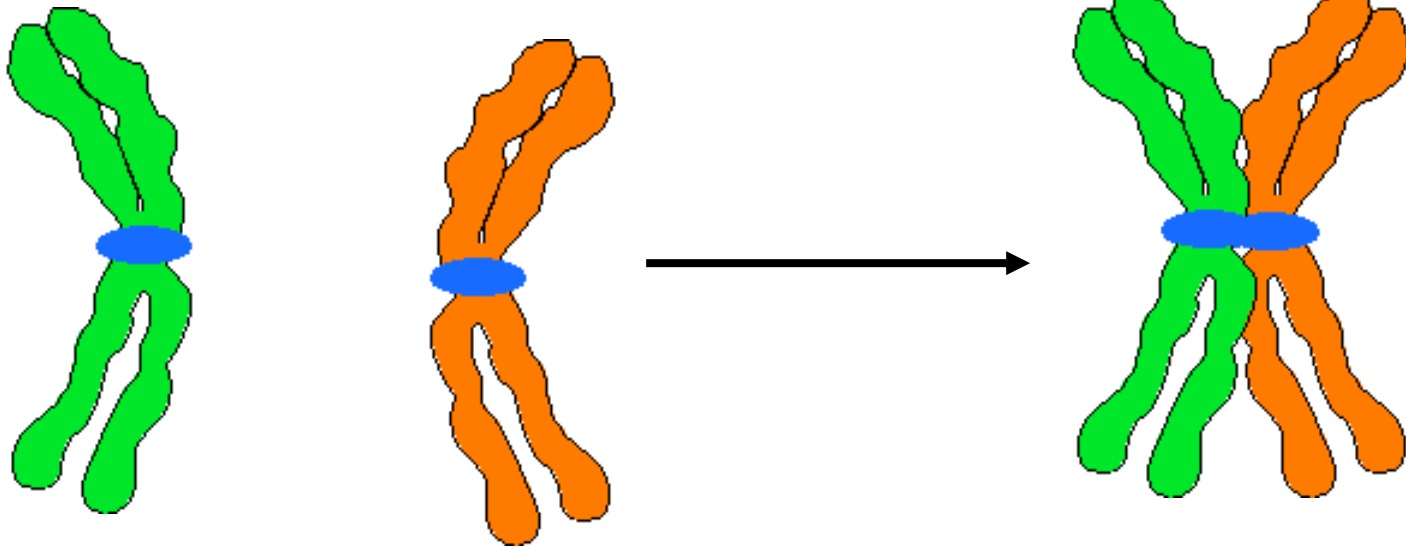
Nuclear
envelope

Telophase I
(diploid)

Tetrads Form in Prophase I

Homologous chromosomes
(each with sister
chromatids)

Join to form a
TETRAD



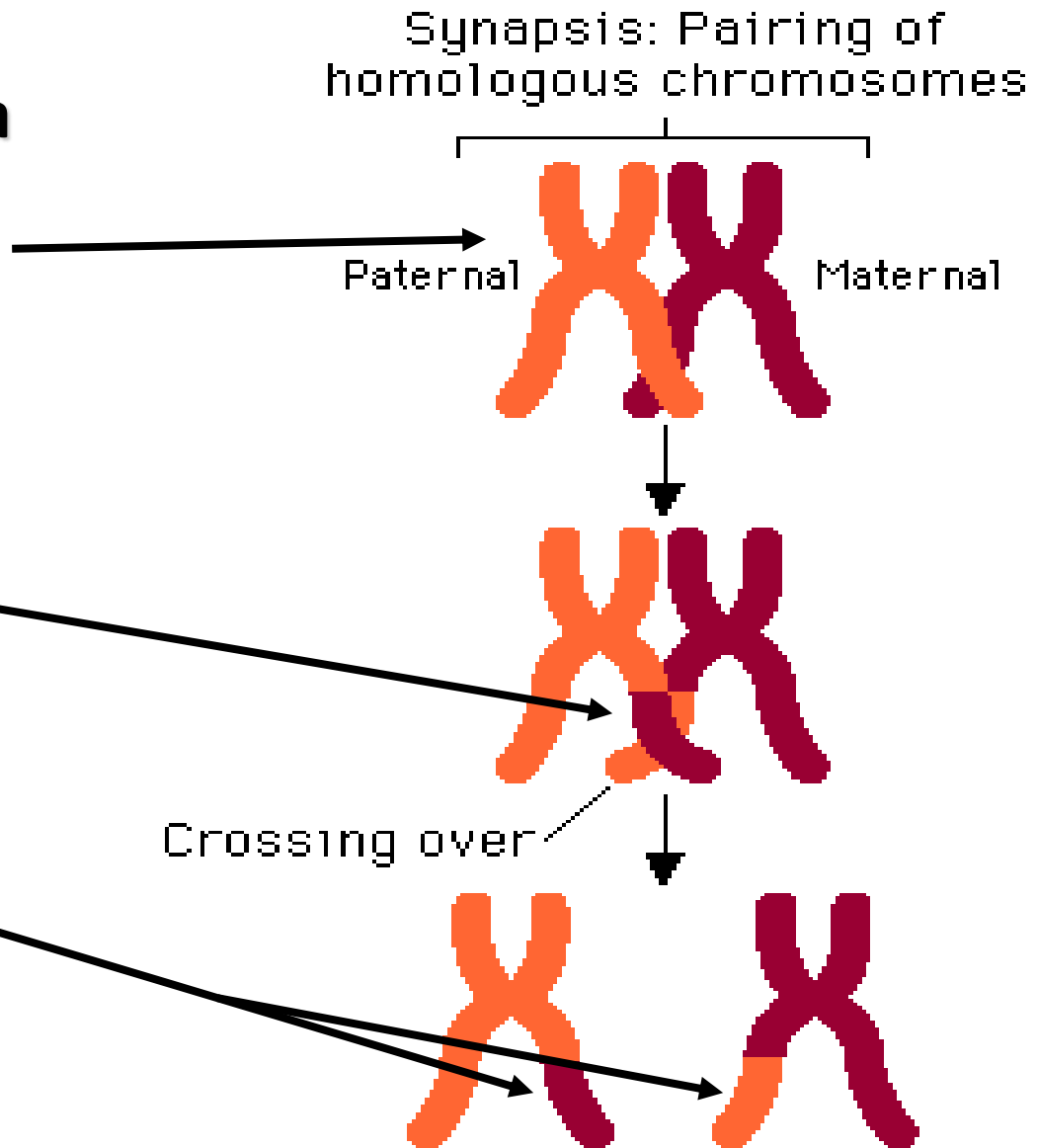
Called Synapsis

Crossing-Over

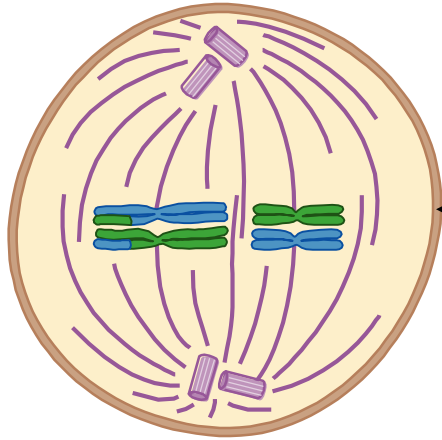
✓ **Homologous chromosomes in a tetrad cross over each other**

✓ **Pieces of chromosomes or genes are exchanged**

✓ **Produces Genetic recombination in the offspring**

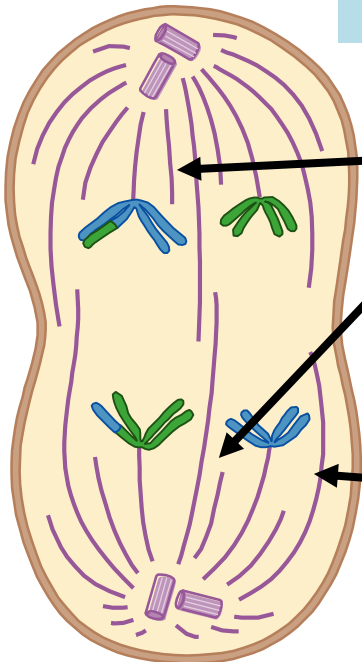


Metaphase I



Homologous pairs of chromosomes align along the **equator** of the cell

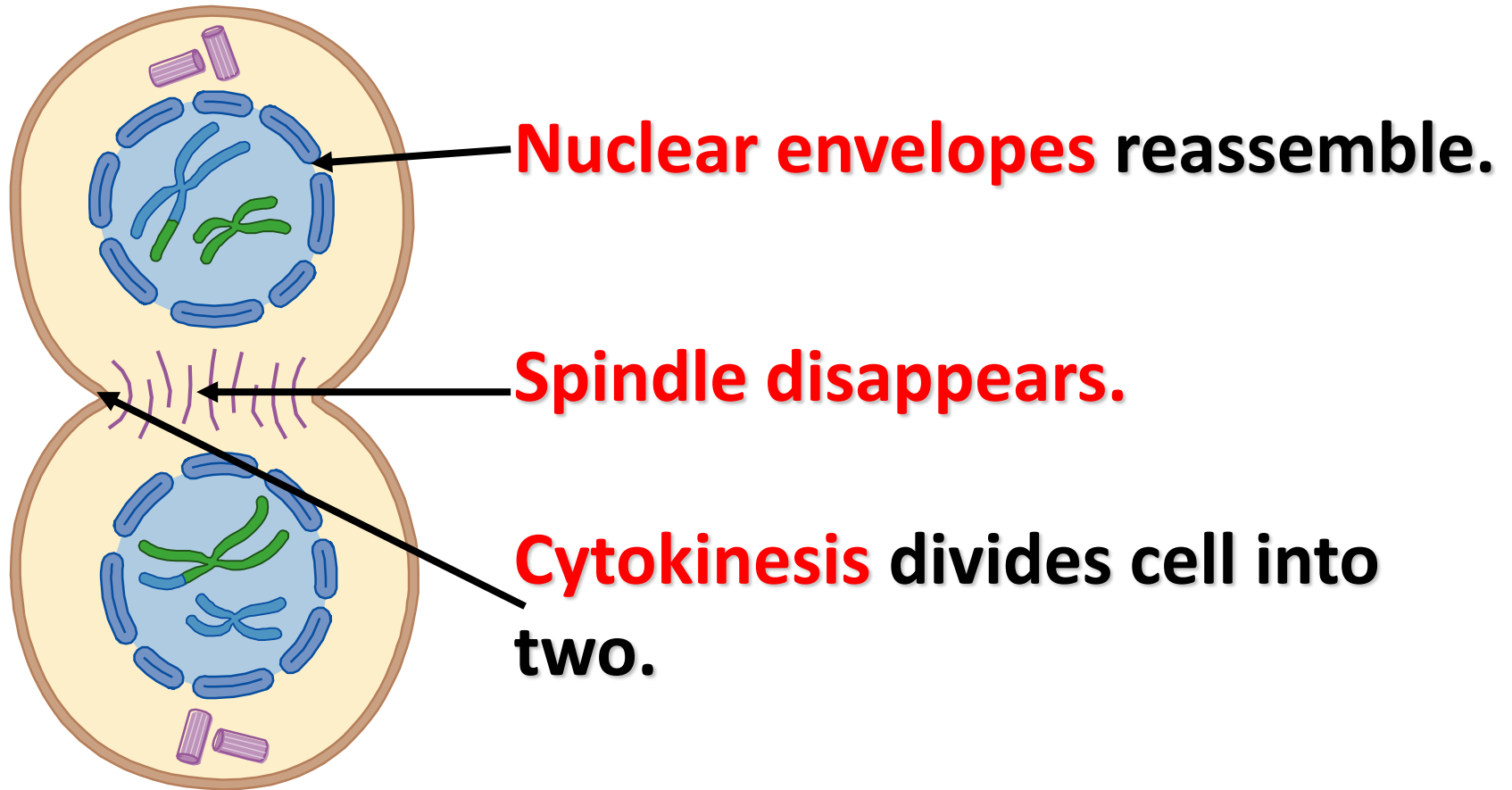
Anaphase I



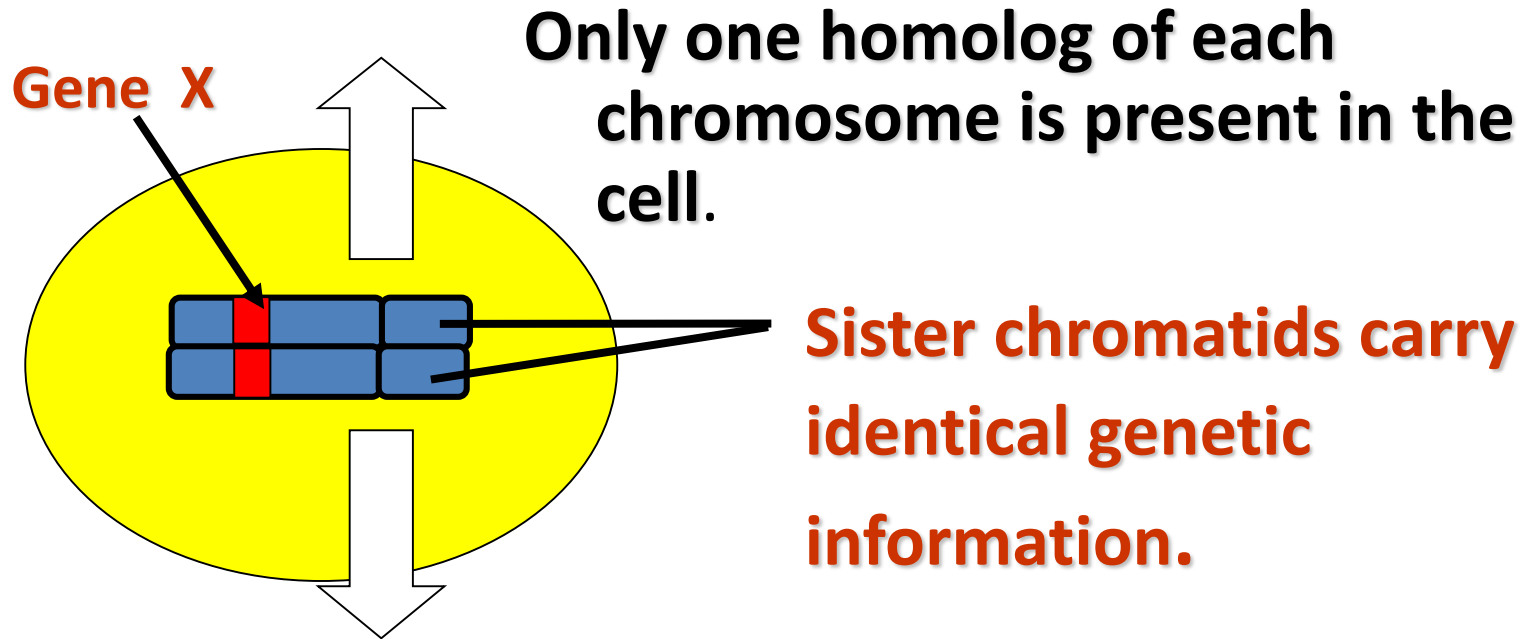
Homologs separate and move to opposite poles.

Sister chromatids remain attached at their **centromeres**.

Telophase I

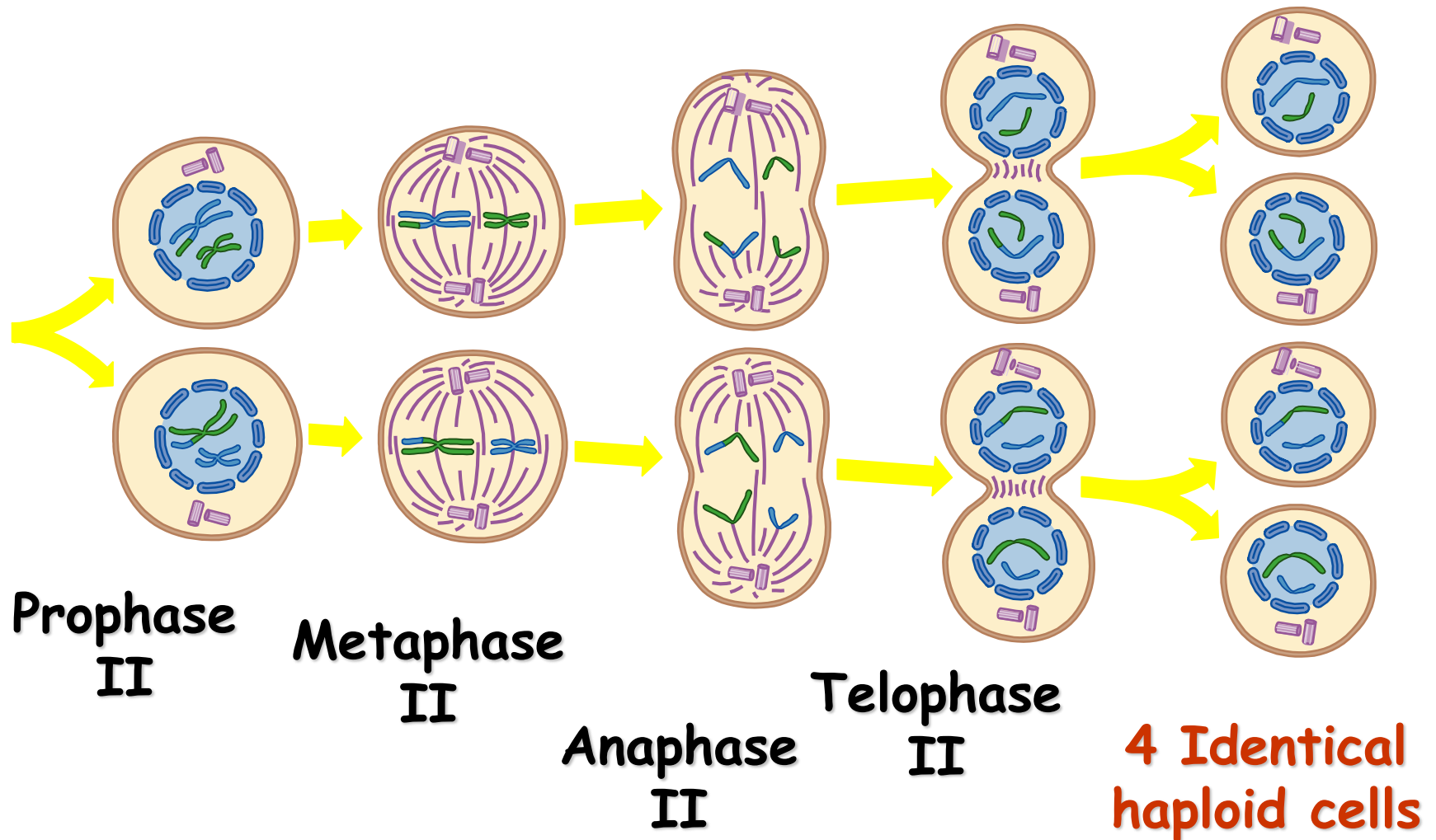


Meiosis II



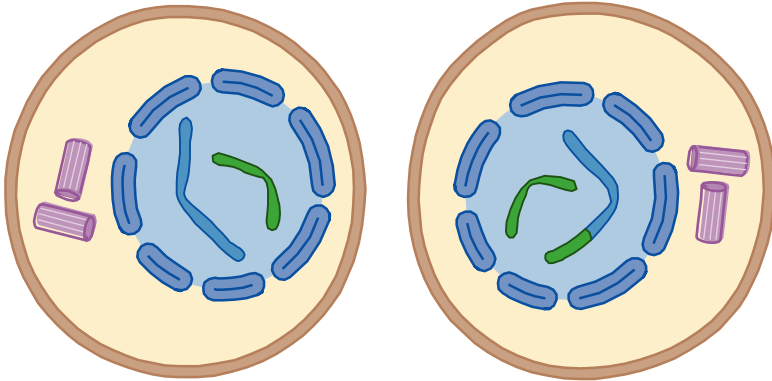
Meiosis II produces gametes with one copy of each chromosome and thus one copy of each gene.

Meiosis II: Reducing Chromosome Number

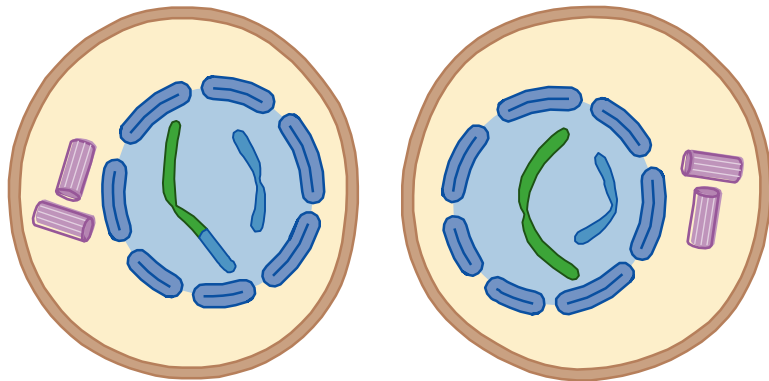


Results of Meiosis

Gametes (egg & sperm)
form



Four haploid cells with
one copy of each
chromosome

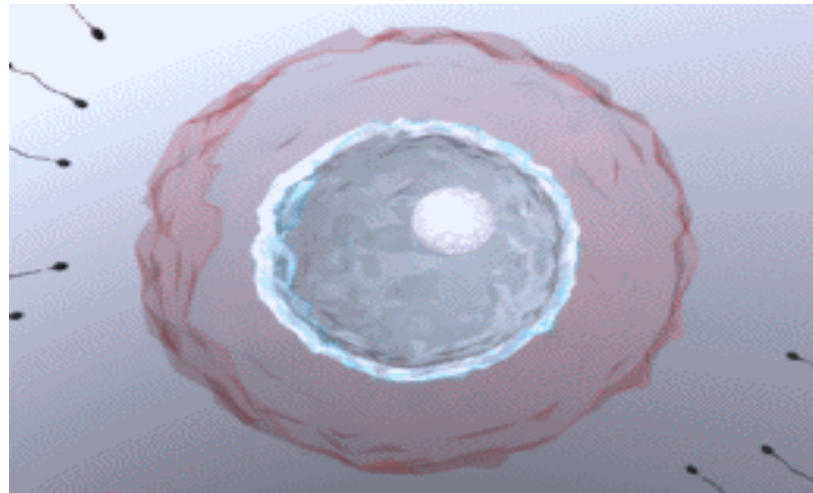


One allele of each gene

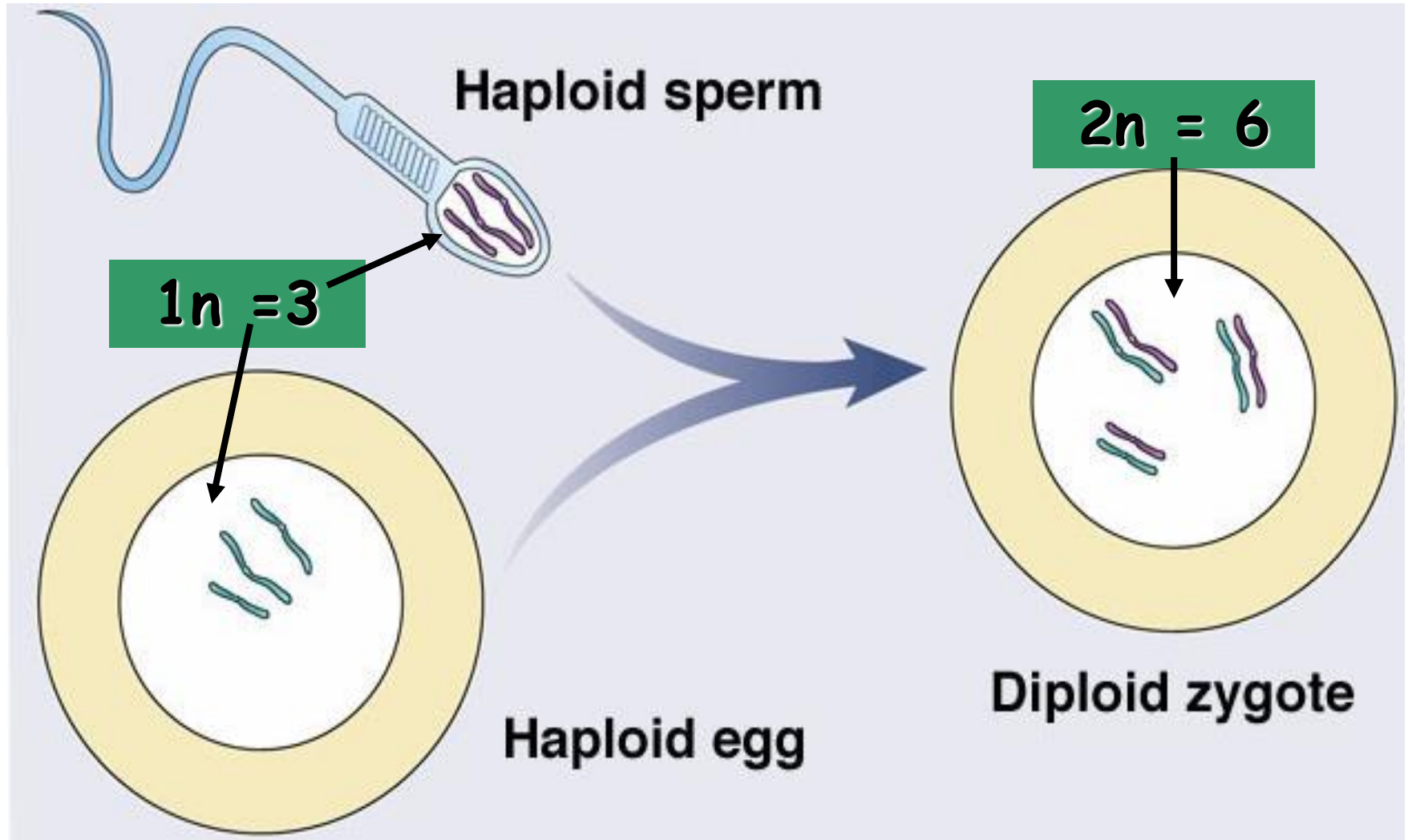
Different combinations
of alleles for different
genes along the
chromosome

Why Do we Need Meiosis?

- ✓ It is the fundamental basis of **sexual reproduction**
- ✓ Two haploid ($1n$) gametes are brought together through **fertilization** to form a diploid ($2n$) zygote

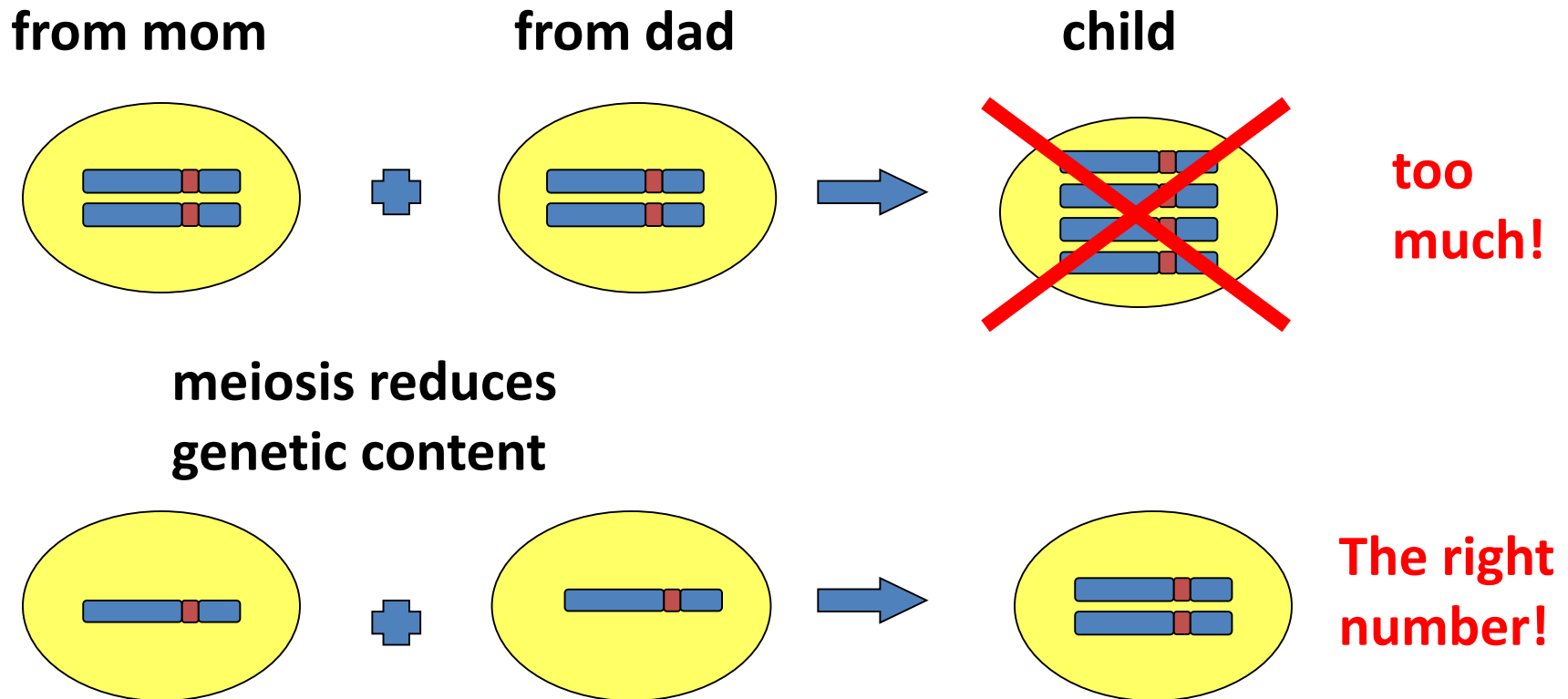


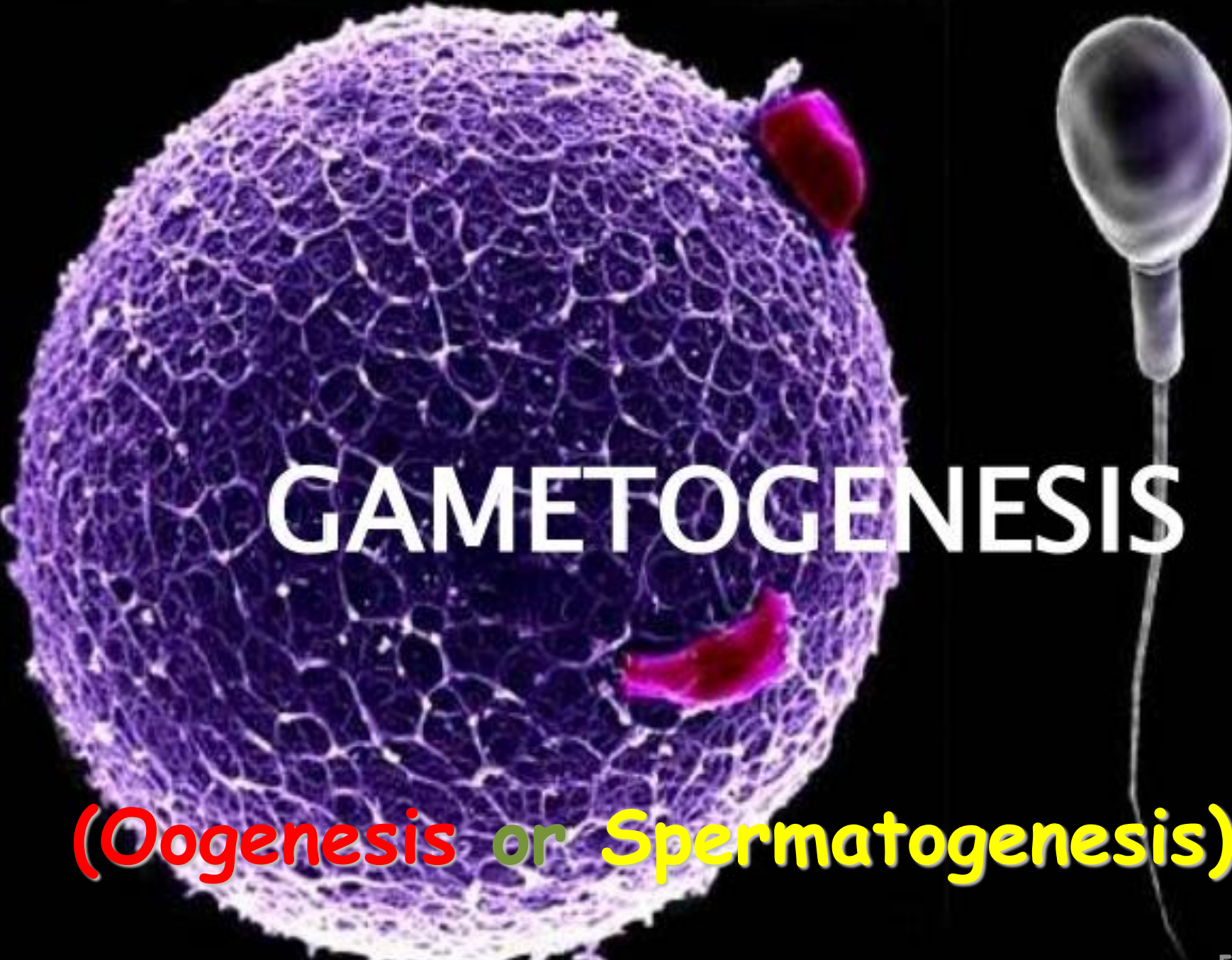
Fertilization - "Putting it all together"



Meiosis Forms Haploid Gametes

- ✓ Meiosis must **reduce the chromosome number by half**
- ✓ **Fertilization then restores** the $2n$ number



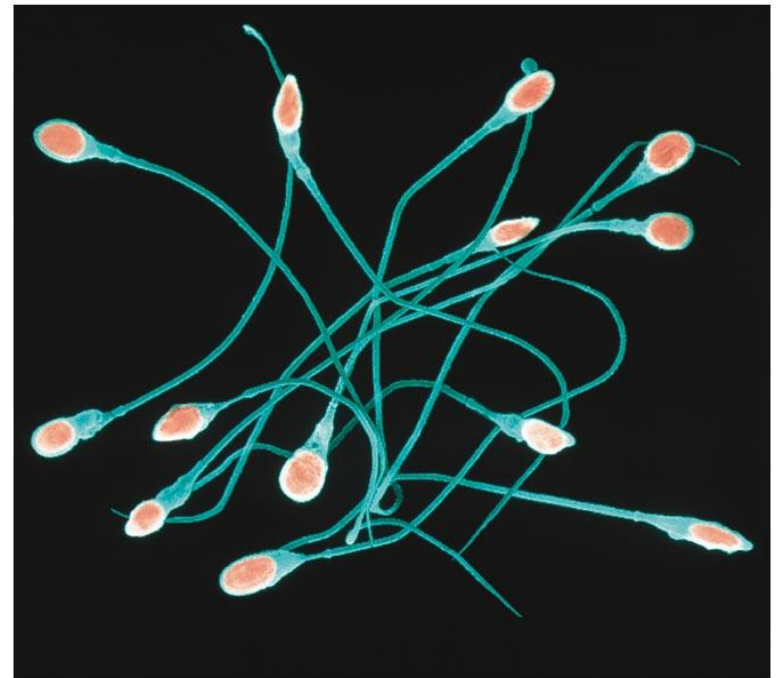
The image features a large, spherical cell on the left with a complex, purple, mesh-like surface texture, representing an oocyte. To its right is a smaller, elongated sperm cell with a light blue, oval head and a thin, white tail. The background is solid black.

GAMETOGENESIS

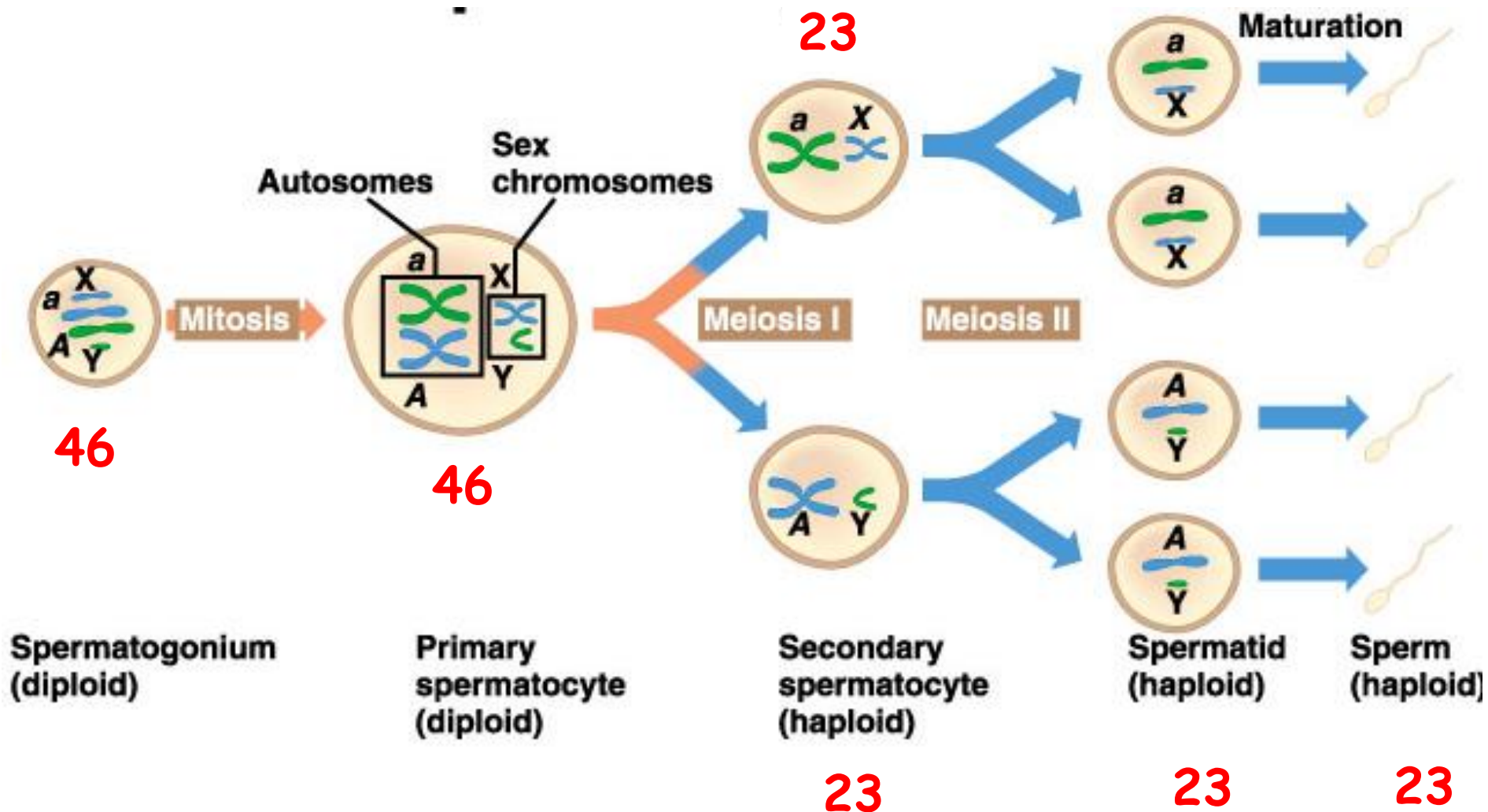
(Oogenesis or Spermatogenesis)

Spermatogenesis

- ✓ Occurs in the **testes**
- ✓ Two divisions produce 4 spermatids
- ✓ Spermatids **mature into sperm**
- ✓ Men produce about **250,000,000 sperm per day**



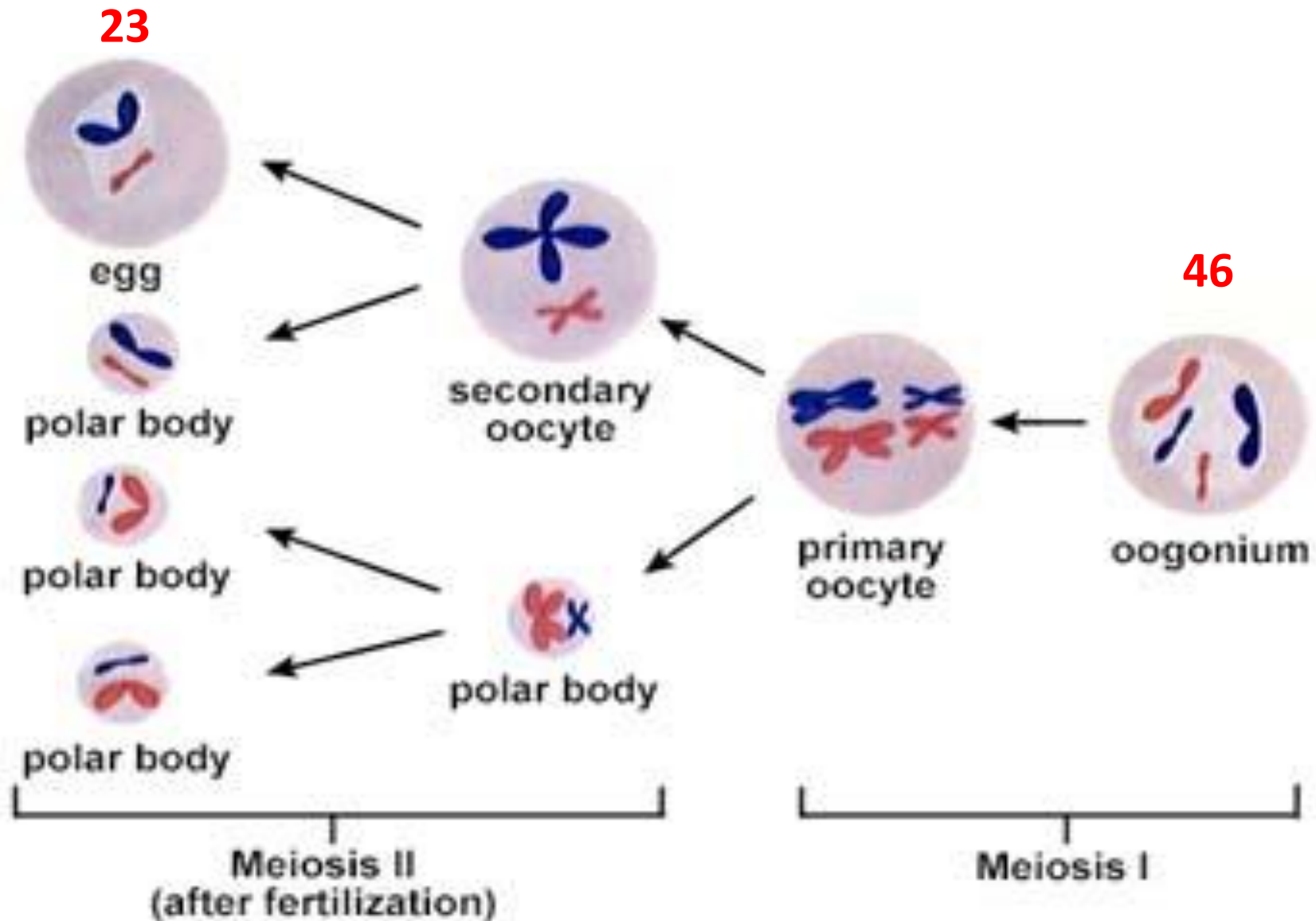
Spermatogenesis in the Testes



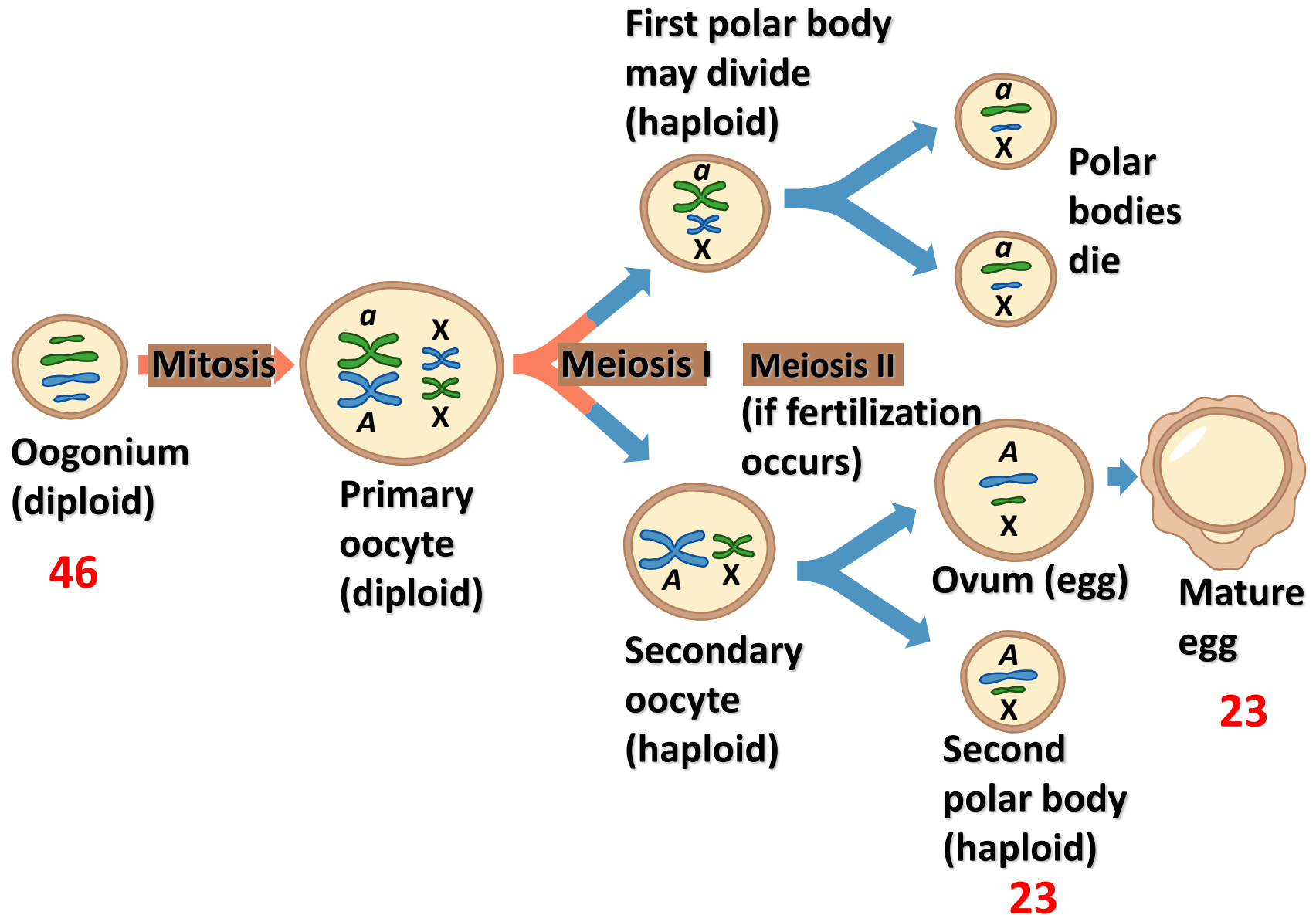
Oogenesis

- ✓ Occurs in the **ovaries**
- ✓ Two divisions produce **3 polar bodies** that die and **1 egg**
- ✓ Polar bodies die because of **unequal division of cytoplasm**
- ✓ Immature egg called **oocyte**
- ✓ Starting at puberty, one oocyte matures into an **ovum (egg)** every 28 days

Oogenesis in the Ovaries



Oogenesis in the Ovaries



Comparison of Divisions

Mitosis

Meiosis

Number of divisions	1	2
Number of daughter cells	2	4
Genetically identical?	Yes	No
Chromosome #	Same as parent	Half of parent
Where	Somatic cells	Germ cells
When	Throughout life	At sexual maturity
Role	Growth and repair	Sexual reproduction

Thank You